BEGINNER'S COMPLETE GUIDE

Review of Literature in Medical Research

Everything explained simply — with real examples, fun analogies, and zero scary jargon.

Search | Read | Judge | Write

Dept. of Community Medicine · AIMS Faridabad · For Beginners

What Is a Literature Review?

Let's start with a simple story...

ANALOGY: Imagine you want to build the world's best LEGO castle. Before you start, you check what castles others have already built, what worked, and what didn't. THAT is a literature review!

What it IS

Searching through all the research that has already been done on a topic, reading it carefully, and summarising the most important findings.

Why doctors do it

Before a doctor starts a new study, they need to know what everyone else has already discovered — so they don't repeat the same work and waste time!

The golden rule

Always check what others know BEFORE you start. A good literature review proves your question is NEW and your research is NEEDED.

> **85% of all medical research money is WASTED because researchers did not check what was already known.**
>
> *— Chalmers & Glasziou, The Lancet (one of the world's top medical journals)*

Think about it this way 👇

⏳ Wasted YEARS

A researcher spent 3 years proving something that had ALREADY been proven 10 years earlier!

💰 Wasted MONEY

Crores of rupees spent on asking a question someone already answered.

🛡 Real patient harm

If doctors use outdated info, patients get the wrong treatment.

Where Does Literature Review Fit?

It's Step 2 in the whole research journey — and you CANNOT skip it!

ANALOGY: Think of doing research like a road trip. The Literature Review is your MAP — you check it BEFORE driving, not after getting lost!

1

Pick a Topic

What do you want to study?

2

Review Literature

☆ THIS IS WHERE WE ARE! ☆

3

Form a Hypothesis

Make an educated guess

4

Collect Data

Run your study or experiment

5

Analyse Results

Use maths and statistics

6

Write & Publish

Share findings with the world

What a Good Literature Review Does For You

Think of it as your SUPERPOWER as a researcher ✨

🗺 Shows you the map

You learn what the world already knows — so you start where others left off, not from zero.

🔍 Spots the gaps

You find the questions that have NOT been answered yet. That's your research opportunity!

🧭 Points your direction

Helps you design your study better — because you learn from others' mistakes and successes.

🎤 Gives you credibility

When you cite other research, people take YOUR research more seriously.

⏱💰 Saves time and money

You don't accidentally study something already solved. Resources go to real unknowns.

📰 Helps you get published

Journals ask: 'What does this add to what we know?' Your lit review answers that question.

Your Research Libraries 🔍

Six amazing places where medical knowledge lives...

Two Types of Sources — Primary & Secondary

🍎 ***ANALOGY: PRIMARY sources are like picking fruit directly from a tree. SECONDARY sources are like reading someone else's review of a fruit shop.***

PRIMARY SOURCES

Research papers written by the ACTUAL scientists who did the experiment. These are the most reliable.

- ✓ Original research articles
- ✓ Clinical trial reports
- ✓ Case studies and case series
- ✓ Theses and dissertations
- ✓ Conference papers (unpublished data)

SECONDARY SOURCES

Someone else READ the primary research and wrote a SUMMARY or REVIEW. Very useful but one step removed.

- ✓ Review articles and meta-analyses
- ✓ Systematic reviews (Cochrane)
- ✓ Textbooks and reference books
- ✓ Government health reports
- ✓ Guidelines (WHO, ICMR, MOHFW)

The 6 Big Medical Libraries Online

These are like Google — but only for medical research papers, and much more reliable!

PubMed

FREE ✓

The BIGGEST medical library in the world. Run by the US government. Has 36 million articles. Always start here.

IN Indexes key Indian journals like IJCM, IJP, JAPI

Cochrane

FREE ✓

Only has SUPER HIGH QUALITY reviews. If Cochrane says something, doctors trust it completely.

IN Has India-focused systematic reviews

Embase

PAID

Enormous database, especially good for medicines and drugs. Covers more European research than PubMed.

IN Best for drug side-effect studies in India

IndMED/NML

FREE ✓

The Indian medical library! Contains research from Indian hospitals and universities that bigger databases miss.

IN IN India's own medical database — don't skip!

Ovid/Scopus

PAID

Helps you find WHO CITED a paper. Great for tracking how ideas spread across research communities.

IN Good for seeing how many researchers cited an article

Google Scholar

FREE ✓

Like regular Google but ONLY for academic papers. Great as a starting point but can find low-quality sources too — be careful!

IN Finds Indian theses, reports, and government data

Types of Literature Reviews — Which Is Which?

ANALOGY: These are like different ways of making a cake — same ingredients (research papers), very different final products!

Systematic Review

BEST quality. Uses a strict checklist. Tells you DEFINITIVELY what the evidence says.

e.g. 'Does handwashing reduce diarrhoea in children under 5?' — one definitive answer.

Narrative Review

A summary of what you read. Less strict. More flexible. Good for broad topics.

e.g. 'Overview of hypertension management in India' — tells the story.

Scoping Review

Used to MAP a topic when you don't even know where to start. Great for new areas.

e.g. 'What research exists on ASHA worker mental health?' — a landscape survey.

Meta-Analysis

Combines numbers from MANY studies using special maths to get ONE big answer.

e.g. Combines 20 studies on a vaccine to get one combined effectiveness percentage.

Umbrella Review

Reviews of systematic reviews! The highest level. Compares multiple top-quality reviews.

e.g. 'What do all the Cochrane reviews say about TB prevention?' — the ultimate summary.

Rapid Review

A quick, streamlined review. Done fast when a decision needs to be made urgently.

e.g. During COVID-19, rapid reviews helped doctors decide treatments within weeks.

Grey Literature — The Hidden Treasure 💎

🏴‍☠️ ANALOGY: Most people only look in bookshops for books. Grey literature is like finding rare books in someone's attic — not in any shop, but incredibly valuable!

Grey literature = anything NOT published in a regular journal, but still containing valuable research data.

IN

Indian Govt Reports

- NFHS-5 health survey data
- MOHFW annual reports
- NHM programme guidelines
- ICMR technical reports

International Bodies

- WHO Global Health Observatory
- UNICEF country reports
- World Bank health data
- IHME Global Burden of Disease

Universities & Theses

- Shodhganga (India's theses portal)
- PhD and MD theses
- University research reports
- Unpublished study data

Trial Registries

- ClinicalTrials.gov (global trials)
- CTRI — India's trial registry
- ISRCTN (international trials)
- WHO ICTRP (all registries combined)

⚠️ IMPORTANT: If you skip grey literature, your review will have GAPS — and reviewers will notice!

PART 3 — HOW TO ASK

Ask the Right Question First

PICO turns a fuzzy hunch into a laser-sharp search...

PICO — Your Magic Question Formula 🧮

🍕 ***ANALOGY: Before ordering pizza, you decide: WHO is eating (P), WHAT topping (I), compared to WHAT else (C), to make everyone HOW happy (O). PICO works the same way for research!***

P

POPULATION

WHO is being studied?

The group of people you are studying

📌 e.g.
Adults over 40 with Type 2 Diabetes in rural Haryana

I

INTERVENTION

WHAT is being done?

The treatment, programme, or thing you're testing

📌 e.g.
ASHA worker foot care programme

C

COMPARISON

Compared to WHAT?

The alternative — usually the current standard treatment

📌 e.g.
Normal hospital outpatient care, nothing special

O

OUTCOME

WHAT is measured?

How you know if it worked or not

📌 e.g.
Number of foot ulcers after 12 months

SECTION B

Beyond PICO

PICOS	**Population · Intervention · Comparison · Outcome · Study Design** *Best for: Systematic reviews where study design is a key inclusion criterion*
PECO	**Population · Exposure · Comparison · Outcome** *Best for: Epidemiology & environmental health — no active intervention involved*
SPIDER	**Sample · Phenomenon · Design · Evaluation · Research type** *Best for: Qualitative and mixed-methods research in community/nursing settings*
ECLIPS	**Expectation · Client group · Location · Impact · Professionals · Service** *Best for: Health services management and health system organisation questions*

Quick rule — Intervention present → PICO/PICOS. Exposure only → PECO. Qualitative focus → SPIDER.

⍰ EXERCISE: Build a PICO Question Step by Step

📋 SCENARIO: You are a doctor in Haryana. You wonder — can zinc tablets help children recover from diarrhoea faster than just ORS alone?

P

Step 1 — Who are the children?

✅ Answer: Children under 5 years old, in India, with diarrhoea

💡 Age? Location? What condition do they have?

I

Step 2 — What treatment are you testing?

✅ Answer: Zinc tablets given along with ORS (oral rehydration salts)

💡 What's the NEW thing you want to test?

C

Step 3 — What's the comparison?

✅ Answer: ORS alone — no zinc tablets, just the standard treatment

💡 What do you compare the new treatment TO?

O

Step 4 — How will you know it worked?

✅ Answer: Number of days until the child has no more diarrhoea

💡 What can be measured? Shorter illness = zinc worked!

Turning Your PICO into Search Words

🌐 *ANALOGY: When you search on Google, you try different words for the same thing — 'car', 'automobile', 'vehicle'. Medical databases need the same trick!*

The secret: DIFFERENT papers use DIFFERENT words for the same thing. You need ALL the words!

P	**Children with diarrhoea in India**	Search words to use: "Children" "Pediatric" "Under five" "Infant" "India" "South Asia"
I	**Zinc supplementation**	Search words to use: "Zinc" "Zinc sulfate" "Zinc gluconate" "Zinc therapy" "Zinc tablets"
C	**ORS / standard care**	Search words to use: "ORS" "Oral rehydration" "Standard treatment" "Control group"
O	**Duration of diarrhoea**	Search words to use: "Diarrhoea" "Diarrhea" "Gastroenteritis" "Stool frequency" "Recovery time"

Search Like a Detective 🔎

AND, OR, NOT — three magic words that control everything...

Build Your Search Strategy — Six Steps Before You Type

1 Identify Core Concepts from PICO

Extract 2–4 concepts. Each concept becomes a separate block. Do NOT combine different concepts yet

2 Generate Synonyms for Each Concept

'Key terms vary across time.' For each concept, list all synonyms, acronyms, and spelling variants .

3 Find the MeSH Equivalent

Map each concept to its controlled vocabulary term at mesh.nlm.nih.gov before writing the search.

4 Combine with Boolean Operators

OR links synonyms within one concept. AND connects different concepts.

5 Apply Limits and Filters

Narrow by: year of publication, document type, discipline, location, or database AFTER building core search

6 Document and Save the Strategy

Record date, database, exact string, hit count. 'Walk students through how to save their search and receive alerts.'

The 3 Magic Search Words — AND · OR · NOT

⍰ ANALOGY: These three words are like volume knobs on a music system. AND makes the search quieter (fewer results), OR makes it louder (more results), NOT mutes certain sounds.

AND

Makes results SMALLER

Both words must appear in the same article.

```
"Zinc" AND "Diarrhoea" → only articles that talk about BOTH
```

Use AND to combine DIFFERENT ideas

OR

Makes results BIGGER

Either word can appear. Gets all the synonyms.

```
"Diarrhoea" OR "Diarrhea" → gets both spellings!
```

Use OR for SAME idea, different words

NOT

Removes unwanted results

Excludes articles that contain the second word.

```
"Malaria" NOT "Plasmodium" → removes some malaria types
```

Use NOT to REMOVE things you don't want

SECTION C

MeSH Terms — The Master Key Every Expert Uses First

What is MeSH?

Medical Subject Headings (MeSH) is the controlled vocabulary PubMed uses to index every article. 'Because of such a huge collection, it is necessary to acquire skills for searching the relevant articles with minimal effort.' [B1 p.30]

MeSH maps all synonyms to one standardised term — capturing every relevant paper regardless of what words the authors used.

Why you must use it:

✓ Captures papers using any synonym for the same concept

✓ "Tuberculosis"[MeSH] includes TB, Koch's disease, phthisis

✓ Explodes automatically to include all narrower sub-terms

✓ Required in Methods section of published systematic reviews

MeSH in Action — Book Examples

Keyword: "community health workers"

→ **MeSH: "Community Health Workers"[MeSH]**

Includes ASHAs, CHWs, lay health workers, village health visitors

Keyword: "high blood pressure"

→ **MeSH: "Hypertension"[MeSH]**

All subtypes: essential, secondary, resistant, gestational

Keyword: "under five diarrhoea"

→ **MeSH: "Diarrhea, Infantile"[MeSH]**

Explodes: gastroenteritis, cholera infantum, summer diarrhoea

[B1 p.30] 'You may like to consult various online tutorials for PubMed search.' Start at

Deep Learning Example — Build a Complete PubMed Search Live : Do it yourself

Research Question: Does ASHA-led community hypertension screening improve early detection in rural Haryana? (PICOS)

1 PICO CONCEPTS

P: Hypertension · I: Community screening / ASHA · C: Routine care · O: Early detection · S: RCT/Cohort

2 MeSH EQUIVALENTS

"Hypertension"[MeSH] · "Community Health Workers"[MeSH] · "Mass Screening"[MeSH] · "Rural Health"[MeSH]

3 ADD SYNONYMS + OR

("Hypertension"[MeSH] OR "High Blood Pressure"[TIAB] OR "HTN"[TIAB]) AND ("Community Health Workers"[MeSH] OR "ASHA"[TIAB] OR "lay health worker"[TIAB])

4 ADD GEOGRAPHY

AND ("Rural Health"[MeSH] OR "rural India"[TIAB] OR "Haryana"[TIAB] OR "village"[TIAB])

5 APPLY FILTERS

Filters: Humans · English · 2010–2024 · India[Affiliation] · Publication type: RCT, Cohort, Cross-sectional

'After using the same search functions, filters, and key terms, compare results. Direct students to follow your search and then perform the same search on their device.' : Consult online tutorials for PubMed search.

From 1,847 Papers to 41 — How?

PRISMA shows exactly how we sieve out the right studies...

PRISMA — How Doctors Pick the Right Papers 🎭

🎭 ***ANALOGY: Imagine 1,847 kids audition for your school play. PRISMA is the 4-round audition process — in each round, more kids get eliminated until only the BEST 41 make the final cast!***

All papers found in databases **n = 1,847**

After removing duplicates **n = 1,240**

✕ 607 duplicates removed — same paper in 2 databases!

After reading titles & abstracts **n = 348**

✕ 892 removed — clearly off topic

After reading full paper **n = 41**

✕ 307 removed — wrong design or missing data

Inclusion & Exclusion Criteria — Your Bouncer at the Door 🚪

🏪 *ANALOGY: Like a shop that only sells children's shoes — any shoes that are not children's shoes get EXCLUDED. You decide the rules BEFORE shopping, not while shopping!*

✅ INCLUDE (Let IN)

- Study design: RCTs, cohort, or cross-sectional surveys
- Population: children under 5, in India
- Intervention must involve zinc supplementation
- Outcome measures diarrhoea duration or frequency
- Published between 2010 and 2024

❌ EXCLUDE (Keep OUT)

- Animal studies (we need human/ studies in children!)
- Studies from outside India (different context)
- No control/comparison group included
- Only available as a conference abstract (no full paper)
- Editorials, letters, or opinion pieces

★ *GOLDEN RULE: Always decide your criteria BEFORE you start screening. Never change the rules mid-game!*

What to Extract From Each Paper

ANALOGY: Think of each paper like a fruit. Data extraction is like squeezing out the JUICE — you want just the key information, not the whole fruit.

Paper Identity

- Author names & year published
- Journal name & impact factor
- Country where study was done
- Was it funded? By whom?

Who was studied

- How many people (sample size)?
- Age, sex, background of participants
- Where were they (rural/urban/hospital)?
- Inclusion & exclusion criteria used

The Intervention

- Exactly what treatment was given?
- Dose, duration, frequency
- Who delivered the intervention?
- What was the control group given?

The Results

- Main findings (numbers with confidence intervals)
- Was the result statistically significant?
- What was the effect size?
- Any side effects or harms reported?

Study Quality

- Was randomisation done properly?
- Were groups comparable at baseline?
- Was blinding used to reduce bias?
- Overall risk of bias rating (low/high)

Conclusions

- What did the AUTHORS conclude?
- What did YOU (the reviewer) conclude?
- Does this fit or contradict other studies?
- What questions does this paper leave open?

PART 6 — HOW TO JUDGE

Don't Believe Everything! ⚖️

Published does NOT mean proven — here's how to check...

Critical Appraisal — Judge Every Paper!

ANALOGY: Just because something is printed in a book doesn't make it TRUE. A critical appraisal checklist is your lie-detector test for medical research!

Ask these 7 questions about EVERY paper you want to include in your review:

1	**Is the question clear?**	Is the research question specific enough? Vague questions = vague answers.
2	**Is the design right?**	Does the type of study match the question? (RCT for treatment questions, etc.)
3	**Was it big enough?**	Was the sample size large enough? Small studies can get lucky results by chance.
4	**Were groups the same?**	At the start, were the two groups (treatment vs control) actually comparable?
5	**Were measurements valid?**	Did they measure the right thing, in the right way, with a reliable tool?
6	**Was the stats correct?**	Were the right statistical tests used? Was p-value confused with importance?
7	**Can I trust the result?**	Do the CONCLUSIONS match what the DATA actually showed? Any hidden bias?

Bias — The Invisible Enemy of Research 😵‍💫

⚖️ ANALOGY: Bias is like a wonky scale — even if you weigh things correctly, the wrong answer keeps coming out. Bias makes research results look different from reality.

📰 Publication Bias

Only POSITIVE results get published. Negative results sit in a drawer. So the published evidence looks MORE positive than reality!

Fix: Search trial registries for unpublished studies too.

🎯 Selection Bias

The group of people chosen for the study doesn't represent normal patients. E.g. only wealthy hospital patients, not rural villagers.

Fix: Check if the study population matches YOUR patients.

🔤 Language Bias

Researchers who only read English miss studies published in Hindi, Chinese, German etc. They might miss important evidence!

Fix: Search multiple databases. Try non-English sources.

⏱️ Time-lag Bias

Positive studies get published faster than negative ones. So RECENT reviews look more positive than they should.

Fix: Include preprints and trial registries for fresh data.

🔄 Cherry-picking

A researcher only includes studies that AGREE with their opinion. This is cherry-picking — deliberately misleading!

Fix: Include ALL relevant studies, even ones that disagree.

💎 Definition Bias

Different studies define 'diarrhoea' differently — some say 2 loose stools, others say 3. Results can't be compared!

Fix: Check exact definitions carefully before combining data.

Which Evidence Should You Trust Most?

ANALOGY: Evidence is like sports medals. Gold is the best (systematic reviews), silver is very good (RCTs), bronze is okay (observational studies), and participation ribbon is least reliable (expert opinion).

1 Systematic Reviews + Meta-Analyses	STRONGEST	*Many studies combined. BEST possible answer. Cochrane reviews are here.*
2 Randomised Controlled Trials (RCTs)	VERY STRONG	*Randomly allocate people to groups. Gold standard for treatment questions.*
3 Cohort & Case-Control Studies	STRONG	*Observational studies. Follow people over time. Good for causes.*
4 Cross-Sectional Surveys	MODERATE	*Snapshot in time. Shows prevalence. Cannot prove cause and effect.*
5 Case Reports / Case Series	WEAK	*Interesting stories about individual patients. Cannot be generalised.*
6 Expert Opinion / Editorials	WEAKEST	*What famous doctors think. Can be biased. Never rely on this alone.*

Rule: Always prefer the HIGHEST level of evidence available. If a Cochrane review exists on your topic — start there!

PART 7 — FINISH LINE

Organise, Write & Remember 🏁

Reference managers, the Candy Sort, 6 common mistakes and your 8 golden rules...

Never Lose a Paper Again — Reference Managers

ANALOGY: Without a reference manager, organising your papers is like keeping a MESSY drawer of notes. With one, it's like having a PERFECT filing cabinet — find any paper in 5 seconds!

Zotero

FREE

The most popular free tool. Install the browser plugin — it captures reference details automatically when you're on PubMed. One click saves the paper!

→ Best choice for beginners. Start here.

Mendeley

FREE

Great for reading and highlighting PDFs. Stores the full paper AND your notes together. Like a digital reading desk.

→ Best if you read a lot on screen.

EndNote

PAID

The professional standard. Used for large systematic reviews with 500+ papers. Very powerful. Ask if your institution has a free licence.

→ Check if AIMS has a free licence. No it does have

The Candy Sort — Organise Themes Not Papers!

Write each paper's key finding on a sticky note. Group them by theme. Name each theme. That becomes your review structure!

SECTION H

Six Mistakes That Sink Literature Reviews — Avoid Every One

01

Single Database Only

Fix: Always ≥3 databases. : MedLine, Embase, Cochrane for medical topics. One source misses 30–50% of relevant literature.

02

No Search Documentation

Fiix: 'Walk students through how to save their search and receive search alerts.' Record date, database, string, hit count every session.

03

Cherry-Picking Evidence

Fix: The discussion should clearly resolve any conflicts with existing literature.' Include contradictory studies — it's science, not advocacy.

04

Ignoring Theses & Abstracts

Fix: 'Do not forget to consult previous theses on your topic. Conference abstracts may have to be searched manually.'

05

Citing Review Articles Only

Fix: 'Not all review articles have the same quality. Sieve good from not-so-good ones.' Go to primary sources every time.

06

'Book Report' Format

Fix: Avoid the book report format (summary per paper). Organise by THEMES. 'Scholars must say something new while connecting to what's already been said.'

Your 8 Golden Rules to Remember 🏆

1. ALWAYS search BEFORE designing your study — not after!
2. Search at least 3 databases — PubMed alone misses 30-50% of papers.
3. Use MeSH terms — they find papers that keywords alone will miss.
4. Record EVERY search — database, date, exact words, how many results.
5. Don't skip grey literature — India's real data is often hidden there.
6. Decide IN/OUT criteria BEFORE you start reading. No changing rules!
7. Question every paper — Published ≠ Proven. Use your checklist.
8. Organise by THEMES not by papers — tell a story, not a list.

"A researcher who cannot find evidence cannot generate it. Master the search — then master the world." 🌍

What Does 'Lacuna' Mean?

One word. Enormous meaning. Let's decode it.

> ***ANALOGY: Imagine a completed 1,000-piece jigsaw puzzle — but 3 pieces are MISSING. You can see the picture, but those blank spots drive you crazy. Those missing pieces are LACUNAE in research!***

The Latin word

Lacuna comes from Latin, meaning a PIT, a HOLE, or a MISSING PART. The plural is LACUNAE (lah-KOO-nee). In science, it means a gap in knowledge.

In medical research

A lacuna is an area that has NOT been studied yet, OR has been studied poorly, OR where results conflict and no clear answer exists.

Why it matters

Every new research study must fill a lacuna. If there is no gap, there is no reason to do the study! Gaps JUSTIFY your research question.

The Universe of Knowledge

Think of all human knowledge as outer space...

Explored Territory

Everything humanity has already studied and published. Millions of research papers, textbooks, guidelines.

Lacunae (The Gaps)

Spots inside explored territory where knowledge is MISSING, WEAK, or CONTRADICTORY. Every '?' is a potential research project!

The Unknown Frontier

Areas NOBODY has looked at yet. Not just gaps — entire undiscovered territories. These are the boldest research questions.

Why Lacunae Are GOLD for Researchers

ANALOGY: A gold miner doesn't dig where others have already dug. She finds a FRESH SPOT of ground no one has touched. Lacunae are your fresh spots — untouched ground where you can strike gold!

A lacuna does THREE things for your research career:

1 TICKET TO PUBLISH

Every research journal asks: 'What does this ADD to what we know?' If you can point to a gap and say 'THIS answers THAT', your paper has a chance of being accepted.

Real example: 'No study has examined ASHA-led hypertension screening in peri-urban Haryana. THIS study fills that gap.'

2 TICKET TO FUNDING

ICMR, WHO, Wellcome Trust — every funding body asks: 'Why is this research needed NOW?' Identifying a lacuna proves the need exists and the investment is justified.

Real example: 'Indian NFHS-5 data shows 28% of diabetics are undiagnosed — yet no intervention study exists for rural PHC settings.'

3 TICKET TO YOUR DEGREE

For MD/PhD theses, the examiners will always ask: 'Why THIS topic?' Pointing to a specific, named lacuna shows intellectual rigour and research maturity.

Real example: 'Systematic reviews on zinc therapy exclude Indian children under 2 years — a gap this thesis addresses.'

How Do Knowledge Gaps Form? 🌱

Gaps aren't accidents — each has a reason. Know the reason, own the gap.

Why Knowledge Is Always Incomplete

ANALOGY: 500 years ago, maps of the world had blank spaces labelled 'Here Be Dragons'. Every generation of explorers filled in more blanks — but new blanks always appeared at the edges. Research works the same way!

Science advances

Every answer creates new questions. When we discovered antibiotics, new questions about resistance emerged. Every solution births fresh gaps.

New contexts emerge

A treatment proven in the UK may never have been tested in India. Climate changes. New diseases appear. Populations shift. Context constantly creates new gaps.

Knowledge has expiry dates

A study done in 1990 may be outdated in 2024. Guidelines change. Technology improves. Old answers become old gaps when re-examined.

Resources are limited

Not every question can be funded. Rare diseases, marginalised populations, unprofitable treatments — these remain understudied because money goes elsewhere.

Results go unpublished

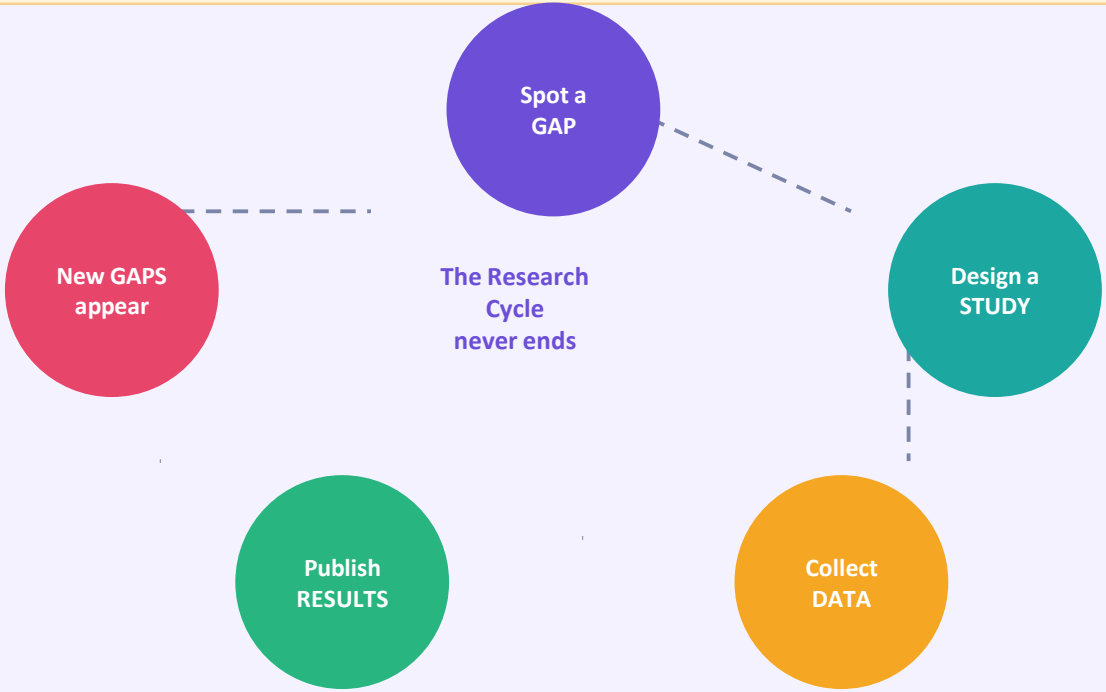
Negative results (treatments that DON'T work) rarely get published. This leaves artificial gaps — we keep repeating failed research because failures are hidden.

Measurement was impossible

Some things couldn't be measured until new technology arrived. Brain imaging, genomics, real-time GPS tracking — technology creates gaps by making new measurements possible.

Every Answer Creates New Questions

ANALOGY: Research is like peeling an onion. You peel one layer, thinking you've reached the centre — and there's ANOTHER layer underneath. Each layer is a gap waiting to be explored!



Key Insight

Researchers who find gaps are not saying previous researchers FAILED.

They are saying: 'You did amazing work — and you opened a door. I will walk through it.'

Gaps are compliments to previous researchers AND springboards for new ones.

IN Real Gaps in Indian Medical Research

These are REAL gaps that exist TODAY — any one of these could become a research project!

Hypertension

What we KNOW:

Prevalence of hypertension in India: well documented nationally.

The GAP:

Effectiveness of ASHA-led blood pressure screening in peri-urban settings of Haryana — NOT studied.

Diabetes

What we KNOW:

Type 2 diabetes rates in urban India extensively researched.

The GAP:

Diabetic foot ulcer prevention by community health workers in rural PHC settings — almost NO data.

Child Nutrition

What we KNOW:

National stunting rates from NFHS-5 well known for all states.

The GAP:

Village-level barriers to ICDS uptake in tribal districts of Jharkhand — rarely studied.

Mental Health

What we KNOW:

Prevalence of depression in general population surveyed multiple times.

The GAP:

Mental health of ASHA workers themselves — their burnout, stress, support needs — barely studied.

Antimicrobial Resistance

What we KNOW:

National antibiogram data collected at tertiary hospital level.

The GAP:

Community-level AMR patterns in rural primary care — a massive, urgent gap.

Geriatric Health

What we KNOW:

Prevalence of chronic disease in elderly people studied at national level.

The GAP:

Implementation of WHO ICOPE screening in Indian field practice areas — not yet evaluated.

How to Spot a Gap in the Literature 🔍

A step-by-step detective method that anyone can learn...

The 5-Step Detective Method

Follow these steps in order. Each step reveals more of the gap.

ANALOGY: Finding a gap is like a detective solving a crime. You gather evidence (read papers), look for contradictions, notice what's MISSING from the crime scene, and name your suspect — the gap!

1 Read the best reviews first

Start with systematic reviews and Cochrane reviews on your topic. These map what IS known and often LIST gaps in their 'Future Research' sections.

2 Check authors' own limitations

Every good paper ends with a 'Limitations' section. These are authors ADMITTING where their study fell short — that's a gap handed to you on a plate!

3 Look at 'Future Research' sections

Authors often explicitly write: 'Future studies should examine...' This is a direct invitation. Accept it! Write it down as a potential lacuna.

4 Ask what's NOT there

After reading 20 papers, ask: 'Which population was never included? Which geography? Which age group? Which comparison was never made?' Absence is a gap.

5 Find the contradictions

When two papers say OPPOSITE things, that's a gap! Conflicting evidence means the question hasn't been properly answered yet — your job is to settle it.

Where Gaps Hide Inside a Paper

ANALOGY: A research paper is a treasure map. The X marking the spot isn't in the middle — it's hidden in specific SECTIONS. Know where to look and you'll find gaps in minutes!

INTRODUCTION

Gap richness: 🔥🔥🔥

The final paragraph ALWAYS states the rationale for the study — i.e. the gap it fills. Copy this sentence. It tells you exactly what the authors saw as missing.

METHODS

Gap richness: 🔥

Inclusion/exclusion criteria reveal WHO was left out. Every excluded group is a potential gap. 'We excluded rural participants' = rural participants = your gap.

RESULTS

Gap richness: 🔥🔥

Subgroup analyses with small numbers say 'we tried but couldn't prove it here'. That's a gap. Tables with missing cells or 'not reported' also signal gaps.

DISCUSSION

Gap richness: 🔥🔥🔥🔥

The RICHEST source. Authors compare to prior work and say 'unlike Study X, we found Y — this discrepancy needs investigation.' = Gap identified!

LIMITATIONS

Gap richness: 🔥🔥🔥🔥🔥

THE GOLDMINE. 'Our sample was small.' 'We could not blind participants.' 'We only followed up for 6 months.' Each limitation is an explicit, author-admitted gap.

FUTURE RESEARCH

Gap richness: 🔥🔥🔥🔥🔥

Sometimes a separate section, sometimes part of the conclusion. Authors EXPLICITLY say: 'Future studies should examine...' — this is a direct invitation to YOU.

The Gap Matrix — A Systematic Way to Find Gaps

1 2
3 4

ANALOGY: Imagine a school attendance register. Some students (populations/settings) are ABSENT from the list. The Gap Matrix is your attendance register for research — you mark who's missing!

Ask FOUR questions about any topic. Each empty cell = a potential lacuna:

	POPULATION *Who has been studied?*	Children ✓ Adults ✓ Elderly ✓ Pregnant women ? Tribal communities ✗
	SETTING *Where has it been studied?*	Hospitals ✓ Urban community ✓ Rural PHC ? Tribal area ✗ Peri-urban ✗
	TIME *When / for how long?*	Short-term (6 months) ✓ 1 year ✓ Long-term (5+ years) ✗ Repeat cross-section ✗
	INTERVENTION *What approaches have been tested?*	Drug therapy ✓ Lifestyle ✓ mHealth apps ✓ ASHA-led programme ✗ Family-based ✗

✓ = Studied ? = Understudied ✗ = Not studied at all (prime lacuna territory!)

Keep a Gap Journal — Never Lose an Insight

ANALOGY: Think of reading papers like sorting Halloween candy. Every time you find something interesting, don't just eat it — put it in a LABELLED BUCKET (your gap journal) so you can find it later!

What to record for each gap you find:

1	**Source:**	Author, year, journal — where did you find this gap?
2	**Gap Statement:**	In ONE sentence: what exactly is missing?
3	**Gap Type:**	Population? Setting? Method? Time? (see Types slide)
4	**Quote:**	The exact sentence from the paper. Paste it verbatim.
5	**Importance:**	Rate 1-5. Why does this gap matter for patients/policy?
6	**Feasible?:**	Can YOU study this? Resources, access, time available?

Example Gap Journal Entry

Source	Ramachandran et al. (2021), Lancet Diabetes
Gap	ASHA-led foot screening not evaluated in rural PHC
Type	Setting gap + Population gap (rural, PHC level)
Quote	"Results may not generalise to rural primary care settings in low-resource states."
Importance	★★★★★ — 26% of India's diabetics are rural
Feasible?	YES — RHTC Kheri Kalan, Faridabad — 74,874 pop.

Six Types of Knowledge Gaps 🔍

Not all gaps are the same. Naming your gap type makes you sound like an expert!

The Six Types of Knowledge Gaps

Every gap you find belongs to one (or more) of these categories. Name it, and you'll write about it perfectly.

1. Evidence Gap

Nobody has studied this at all.

Like a completely blank page in a recipe book — no recipe exists yet!

e.g. ASHA-led mental health first aid in rural Haryana — zero RCTs exist.

2. Population Gap

The study was done, but NOT on your target group.

Like a shoe shop that only stocks adult sizes — children's feet never measured!

e.g. Cochrane reviews on zinc therapy exclude Indian infants under 12 months.

3. Geographic Gap

Studies exist — but only from OTHER countries.

Weather forecast for London won't tell you Haryana's rainfall!

e.g. All AMR studies on UTI are from USA/Europe. India's community data is absent.

4. Methodological Gap

Studies exist, but they used WEAK methods. Better-quality study needed.

Old photos were blurry. Now we have 4K cameras — retake the photo properly!

e.g. HTN prevalence in rural India measured by unreliable self-report — no objective BP data.

5. Temporal Gap

Studies are too old — things have changed since they were done.

Using a 1990 map to navigate 2024 Delhi — new roads exist, old ones are gone!

e.g. NCD burden studies from pre-COVID era miss the massive post-COVID increase in diabetes.

6. Contradiction Gap

Two or more studies give OPPOSITE answers. Someone needs to resolve this.

Two friends both saw the accident — but describe it completely differently. Who's right?

e.g. Half the studies say salt restriction reduces BP in Indians; half say it makes no difference.

Evidence Gap vs Knowledge Gap — What's the Difference? 🤔

ANALOGY: An EVIDENCE gap is when no apple tree exists. A KNOWLEDGE gap is when the tree exists but nobody has counted the apples properly. Both matter. Both need different solutions!

EVIDENCE GAP

Definition:

The topic has never been directly studied. No data exists. Complete absence of research.

Example:

No study has ever measured the impact of yoga on blood pressure in Indian truck drivers.

How to fill it:

Design a NEW original study. You are the pioneer. No prior roadmap.

Urgency:

Highest — science has a blind spot here.

KNOWLEDGE GAP

Definition:

Studies exist but they are insufficient — too small, wrong population, poor methods, outdated, or contradictory.

Example:

20 studies on yoga and BP exist but ALL were done in urban gym-going adults. Truck drivers excluded.

How to fill it:

Build on existing work with better design, larger sample, or different population.

Urgency:

Very high — we THINK we know but we're probably wrong.

Not All Gaps Are Equal — Strong vs Weak Gaps

⚖️

⚖️ ***ANALOGY: Not all needs are equally urgent. 'We need more research on what colour stethoscopes doctors prefer' is a gap — but nobody cares. 'We need data on which drug prevents maternal deaths in India' is a gap the WORLD cares about!***

WEAK GAP ✗

- ✗ Something trivial or irrelevant to real patients
- ✗ Already studied thoroughly — you just didn't search hard enough
- ✗ Too narrow to yield generalisable findings
- ✗ Driven by your curiosity alone, not by patient need
- ✗ Cannot be studied with available resources or ethics

📌 *'No study has examined the snack preferences of cardiologists in Chandigarh.' — TRUE gap, zero importance.*

STRONG GAP ✓

- ✓ Addresses a real patient problem or policy question
- ✓ Supported by multiple authors in multiple papers
- ✓ Has clinical, public health, or social significance
- ✓ Feasible: can be studied ethically and practically
- ✓ Has clear implications if answered — will change practice

📌 *'No study has evaluated ASHA-led diabetic foot screening in rural Haryana PHCs despite 26% undiagnosed rate.' — Strong, important, feasible.*

Turning Gaps Into Your Research 🚀

From 'I found a gap' to 'I have a research proposal' in four steps...

From Gap → Research Question → Study

ANALOGY: A gap is a river you want to cross. A research question is the DESIGN of the bridge. Your study is the actual BRIDGE being built. No river (gap) = no bridge needed = no study justified!

1	**IDENTIFY THE GAP**	'No study has evaluated ASHA-led diabetic foot screening in rural PHC settings in India.'	*Clear, specific, named gap with evidence that it exists (citations from papers that admit it).*
2	**ARTICULATE THE NEED**	'26.3% of diabetics in rural Haryana are undiagnosed. Foot ulcers cost ₹2.4L per case. ASHA contact rate is 94%.'	*Why does this gap MATTER? Data that shows the size and urgency of the problem.*
3	**FORM THE QUESTION**	'Does ASHA-led diabetic foot screening improve early ulcer detection rates compared to standard care in rural Haryana PHCs?'	*The PICO question that directly addresses the gap. Every word in this question came from Step 1.*
4	**DESIGN THE STUDY**	Cluster RCT: 4 PHCs randomised. ASHA-led screening vs standard care. Primary outcome: foot ulcer incidence at 12 months.	*The study design flows directly from the question. The gap justified every design choice.*

Gaps + Funding — How Lacunae Win Grants 💰

💰 ***ANALOGY: Asking for money without a gap is like opening a shop and saying 'give me rent money' without explaining what the shop sells or why the town needs it. Gaps are your BUSINESS CASE for research funding!***

What ICMR / Funders Ask:

1. Why is this research NEEDED right now?
2. What will this ADD to existing knowledge?
3. Why hasn't someone already answered this?
4. Why is YOUR setting the right place to study this?
5. What is the BURDEN of disease if this gap is not filled?

How Lacunae Answer Them:

- ✓ Cite 3 papers that admit this gap in their limitations section.
- ✓ State: 'This study is the FIRST to examine X in setting Y.'
- ✓ Explain: resource constraints, technology limits, neglected population.
- ✓ Use local burden data (NFHS-5, HMIS, your own hospital records).
- ✓ Calculate: cost of inaction. Lives lost. Money wasted. Use numbers.

How to Write Lacunae in Your Thesis ✍️📖

Exact formulas, sentence starters, and what NOT to say...

Where Gaps Appear in Your Thesis

ANALOGY: Writing about gaps is like building a house. The FOUNDATION (introduction) needs the gap first. The WALLS (literature review) build the case. The DOOR (rationale) opens with the gap as the key!

1 INTRODUCTION
(Last paragraph)

State the gap DIRECTLY. This is the climax of your introduction — everything before leads here.

"'Despite extensive research on X in urban settings, no study has examined X in rural PHC contexts in India — a gap this study addresses.'"

2 REVIEW OF LITERATURE
(Final summary)

After reviewing all existing studies, write a synthesis paragraph that explicitly names the gaps revealed.

"'A review of the literature reveals three key lacunae: (1) absence of Indian data, (2) no studies on ASHA delivery, (3) no long-term outcomes measured.'"

3 RATIONALE
(Standalone section)

Make the gap the ENTIRE rationale. This section exists only to justify your study's existence.

"'The lacuna identified in the literature — the absence of community-level intervention data in rural Haryana — forms the scientific basis for this investigation.'"

4 DISCUSSION
(Future research)

Your study will create NEW gaps. Name them. Researchers who read your work need to know where to go next.

"'This study was limited to adults over 40; future research should examine the 30-40 age group where early intervention may be most impactful.'"

Sentence Starters for Writing Gaps ✍

These formulas work every time. Fill in the blanks for your own topic!

Introducing the gap

"Despite extensive research on [TOPIC], no study has examined [YOUR SPECIFIC ANGLE] in [YOUR SETTING/POPULATION]."

Naming the gap

"A notable lacuna in the existing literature is the absence of data on [TOPIC] among [POPULATION] in [SETTING/COUNTRY]."

Gap from contradiction

"The literature presents conflicting evidence on [TOPIC], with [AUTHOR A] reporting [RESULT 1] while [AUTHOR B] found [RESULT 2]. This discrepancy warrants further investigation."

Gap from limitation

"[AUTHOR] acknowledged that their study was limited to [SETTING/POPULATION], suggesting that findings may not generalise to [YOUR SETTING] — the focus of the present study."

Geographic gap

"Most available evidence originates from high-income countries. Data from India, particularly from [STATE/REGION], remain scarce, representing a significant gap this study addresses."

Methodological gap

"Previous studies have relied on [WEAK METHOD, e.g., self-report]. No study has employed [STRONGER METHOD] to objectively assess [OUTCOME] — a methodological gap the present work fills."

Closing the rationale

"The lacuna identified above — [BRIEF GAP RESTATEMENT] — provides the scientific basis and justification for this study."

Creating future gaps

"Future studies should examine [AREA THIS STUDY COULD NOT COVER], as the present work was constrained by [YOUR LIMITATION]."

Do's and Don'ts of Writing Lacunae

ANALOGY: Writing about gaps is like giving a GPS direction. 'Go somewhere north' is bad. 'Turn left on MG Road, go 2km, and the destination is on your right' is PERFECT. Be specific, evidence-based, and directional!

✓ DO THESE 5 THINGS

- ✓ Cite the papers that ADMIT the gap — don't just claim it exists
- ✓ State the gap in ONE clear, specific sentence first
- ✓ Name the gap TYPE (population/geographic/methodological etc.)
- ✓ Quantify the importance — use numbers to show why it matters
- ✓ State how YOUR study fills THIS specific gap

✗ AVOID THESE 5 THINGS

- ✗ Don't say 'more research is needed' — say WHAT research, WHO for
- ✗ Don't claim a gap you can't prove — show the citation trail
- ✗ Don't write three paragraphs — one strong sentence beats ten vague ones
- ✗ Don't exaggerate — 'nobody on Earth has ever...' sounds lazy, not rigorous
- ✗ Don't forget: the gap in your introduction MUST match your research objective

Worked Examples & Golden Rules 🏆

See real gaps in action, then remember the rules forever...

Three Lacunae Written Out — Before & After

See how a vague idea becomes a powerful, citeable lacuna statement in one step.

ASHA workers & Hypertension

Geographic + Population Gap

✗ WEAK: 'More studies on hypertension in India are needed.'

✓ STRONG: "Despite 30 studies on hypertension prevalence in India, no RCT has evaluated ASHA-led screening in peri-urban PHC settings [Singh 2021; Mohan 2022]. This geographic and population gap forms the rationale for the present study."

Zinc Therapy in Children

Population + Geographic Gap

✗ WEAK: 'Zinc supplementation has been studied but not enough in India.'

✓ STRONG: "Cochrane reviews on zinc therapy (Lazzerini 2018) exclude children under 12 months and studies from South Asia. This population gap means evidence cannot be generalised to Indian infants — the target of this investigation."

Community AMR Surveillance

Methodological + Geographic Gap

✗ WEAK: 'AMR is a big problem and India needs more research.'

✓ STRONG: "Existing Indian antibiogram data originate exclusively from tertiary hospital settings [ICMR 2021]. Community-level AMR patterns in rural PHCs — where 70% of antibiotic prescriptions are written — remain wholly uncharacterised, a critical methodological gap."

The 6 Mistakes Students Make with Lacunae ⚠

Every examiner knows these mistakes. Don't let them happen to you.

01

Claiming a gap without evidence

✓ Fix: Every lacuna needs AT LEAST ONE citation — a paper that admits the gap in its limitations or future research section.

02

Being too vague ('more research needed')

✓ Fix: 'More research needed' is never a lacuna. Name: WHAT research, for WHOM, WHERE, using WHAT method, measuring WHAT outcome.

03

Not actually searching thoroughly

✓ Fix: If you didn't search PubMed + IndMED + Cochrane + grey literature, you can't claim a gap. Someone may have studied it — you just didn't look hard enough.

04

Mistaking a Western gap for an Indian gap

✓ Fix: Lots of evidence may exist from the USA or UK. That does NOT mean India has a gap. Check IndMED. Read Indian journals. Search 'India' as a filter.

05

Mixing up gap with hypothesis

✓ Fix: A gap is what's MISSING. A hypothesis is your GUESS about the answer. They are different stages. State the gap FIRST, then the hypothesis AFTER.

06

Finding a gap but not connecting it to your study

✓ Fix: Always write: 'This gap is the rationale for the present study, which specifically aims to...' The gap and your objective must be MIRROR IMAGES of each other.

Your 10 Golden Rules for Lacunae 🏆

1. A lacuna is a GAP in knowledge — what's missing, weak, or contradictory.
2. Every research study MUST fill a lacuna. No gap = no study needed.
3. Find gaps in: Limitations sections, Future Research, Discussion contradictions.
4. Use the Gap Matrix: Population, Setting, Time, Intervention — four ways to spot gaps.
5. Name your gap type: Evidence, Population, Geographic, Methodological, Temporal, or Contradiction.
6. ALWAYS cite who admits the gap. Never claim a gap without a citation.
7. Be SPECIFIC. 'More research needed' is lazy. Name who, where, what, and why.
8. Strong gaps are important, feasible, and supported by burden data or multiple authors.
9. The gap in your introduction must be the MIRROR of your research objective.
10. Your study will create NEW gaps. Name them in your discussion — that's scientific integrity.

"A lacuna is not an embarrassment of science — it is its beating heart. Every gap is an open invitation from the universe to come and look." 🔭

The Foundation of Academic Integrity

10,000+
Papers retracted globally 2023

62%
Retractions due to duplicate/misconduct

~30%
PG theses fail referencing audit

100%
Journals require standardised style

Gives Credit

Acknowledges intellectual contributions of original authors — the cornerstone of research ethics.

Ensures Reproducibility

Allows readers to locate and verify the original sources cited in your work.

Demonstrates Scholarship

Shows depth and breadth of literature reviewed — critical for thesis evaluation.

Avoids Plagiarism

Proper citation protects against academic misconduct proceedings in MCI/NMC audits.

Components Every Reference Must Have

AUTHORS	YEAR	TITLE	SOURCE
Sharma R, Gupta A, Patel N	(2022) or 2022;	Effect of metformin on HbA1c in T2DM	Journal / Book / Website name

VOLUME/ISSUE	PAGES	DOI/URL	ACCESS DATE
Vol. 14, No. 3 14(3):	pp. 234–241 or 234-41	https://doi.org/10.1234/xxxx	[cited 2024 Jan 15]

Assembled Example (Vancouver):

Sharma R, Gupta A, Patel N. Effect of metformin on HbA1c in type 2 diabetes mellitus: a randomised controlled trial. Indian J Med Res. 2022;14(3):234-41. doi:10.1234/ijmr.2022.14.3.234

Which Style for Which Discipline?

01

Vancouver

Used in: Biomedical & Clinical

NEJM, Lancet, BMJ, Indian journals

02

APA 7th

Used in: Psychology, Social Sciences

Psychiatry, Nursing, MPH dissertations

03

Harvard

Used in: General Academic

Public Health, Community Medicine

04

AMA

Used in: American Medical Assoc.

JAMA, AMA journals, US boards

05

NLM / ICMJE

Used in: National Library of Medicine

PubMed, Medline indexed journals

06

MLA

Used in: Humanities / Literature

Medical humanities essays

SECTION 01

Vancouver Style

The Gold Standard for Biomedical Journals

Numeric Citation System — ICMJE Standard

1

In-text citation: Superscript numbers or (numbers) in order of appearance

e.g., "Insulin resistance is a key factor[1] in metabolic syndrome[2]"

2

Author names: Surname first, then initials, no full stops. List up to 6 authors; add 'et al.' after 6th

e.g., Sharma R, Gupta A, Patel N, et al.

3

Journal name abbreviated as per NLM Catalogue (Index Medicus)

e.g., N Engl J Med | Indian J Med Res | Lancet

4

Format: Authors. Title. Abbrev Journal. Year;Volume(Issue):Pages.

Formatted Examples by Source Type

Journal Article	Rajpurkar P, Irvin J, Ball RL, Zhu K, Yang B, Mehta H, et al. Deep learning for chest radiograph diagnosis. PLoS Med. 2018;15(11):e1002686. doi:10.1371/journal.pmed.1002686
Book	Tortora GJ, Derrickson B. Principles of Anatomy and Physiology. 15th ed. Hoboken (NJ): Wiley; 2017. 1248 p.
Book Chapter	Sharma SK. Respiratory system. In: Harrison TR, editor. Harrison's Principles of Internal Medicine. 20th ed. New York: McGraw-Hill; 2018. p. 1890-1945.
Website / Online	World Health Organization. Diabetes [Internet]. Geneva: WHO; 2023 [cited 2024 Jan 10]. Available from: https://www.who.int/news-room/fact-sheets/detail/diabetes
Conference Paper	Gulshan V, Peng L, Coram M, et al. Development of a deep learning algorithm for detection of diabetic retinopathy. Presented at: JAMA; 2016 Nov 29; Chicago, IL.
Thesis / Dissertation	Mehta A. Role of deep learning in early detection of tuberculosis on chest X-ray [MD thesis]. New Delhi: AIIMS; 2022. 145 p.

Citing AI & Deep Learning Publications Correctly

CNN Model

✗ **WRONG:** Esteva et al showed skin cancer detection using AI (2017)

✓ **CORRECT:** *Esteva A, Kuprel B, Novoa RA, Ko J, Swetter SM, Blau HM, et al. Dermatologist-level classification of skin cancer with deep neural networks. Nature. 2017;542(7639):115-8. doi:10.1038/nature21056*

Preprint / arXiv

✗ **WRONG:** Vaswani mentioned transformers in arXiv

✓ **CORRECT:** *Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, et al. Attention is all you need [Preprint]. arXiv. 2017 Jun 12 [cited 2024 Jan 5]:arXiv:1706.03762. Available from: https://arxiv.org/abs/1706.03762*

Dataset Paper

✗ **WRONG:** MIMIC dataset was used (Johnson 2016)

✓ **CORRECT:** *Johnson AE, Pollard TJ, Shen L, Lehman LH, Feng M, Ghassemi M, et al. MIMIC-III, a freely accessible critical care database. Sci Data. 2016;3:160035. doi:10.1038/sdata.2016.35*

SECTION 02

APA 7th Edition

Author–Date System | Psychology & Social Sciences

Author–Date In-Text Citation System

KEY RULES

- In-text: (Surname, Year) or (Surname & Surname, Year)
- Up to 2 authors: cite all; 3+: use 'et al.' from first citation
- Reference list in ALPHABETICAL order by first author surname
- Title: Only first word and proper nouns capitalised (sentence case)
- Journal title: Full name, italicised; Volume number italicised
- DOI formatted as: https://doi.org/xxxxx
- 7th Ed: No place of publication needed for books

FORMATTED EXAMPLES

Journal

Esteva, A., Kuprel, B., & Thrun, S. (2017). Deep learning for skin cancer classification. Nature, 542(7639), 115–118. https://doi.org/10.1038/nature21056

Book

Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep learning. MIT Press.

Website

World Health Organization. (2023). Diabetes. https://www.who.int/news-room/fact-sheets/detail/diabetes

In-text

(Esteva et al., 2017) — first citation
(Sharma & Gupta, 2022) — two authors
(WHO, 2023) — organisation

Citing AI Research in APA Format

Neural Network Study	LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. Nature, 521(7553), 436–444. https://doi.org/10.1038/nature14539 **In-text: (LeCun et al., 2015)**
Conference Proceedings	He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 770–778. https://doi.org/10.1109/CVPR.2016.90 **In-text: (He et al., 2016)**
Software / Algorithm	Chollet, F., & others. (2015). Keras [Computer software]. GitHub. https://github.com/fchollet/keras **In-text: (Chollet et al., 2015)**
Government / Institutional Report	National Health Authority. (2023). AI in healthcare: policy framework (Report No. NHA-AI-2023). Government of India. https://nha.gov.in/ai-report **In-text: (National Health Authority, 2023)**
Preprint	Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., & Houlsby, N. (2020). An image is worth 16x16 words. arXiv. https://arxiv.org/abs/2010.11929 **In-text: (Dosovitskiy et al., 2020)**

SECTION 03

Harvard Style

Author–Date | Public Health & General Academic

AI/DL References in Harvard Format

Journal Article	Topol, E.J. (2019) 'High-performance medicine: the convergence of human and artificial intelligence', Nature Medicine, 25(1), pp. 44–56. doi: 10.1038/s41591-018-0300-7.
Edited Book	Szolovits, P. (ed.) (2019) Artificial Intelligence in Medicine. New York: Routledge.
Book Chapter	Litjens, G. and Ciompi, F. (2018) 'Deep learning in histopathology', in Greenspan, H. (ed.) Deep Learning for Medical Image Analysis. London: Academic Press, pp. 215–246.
Online Report	WHO (2021) Ethics and Governance of Artificial Intelligence for Health. Geneva: World Health Organization. Available at: https://www.who.int/publications/i/item/9789240029200 (Accessed: 10 January 2024).
Conference Paper	Ronneberger, O., Fischer, P. and Brox, T. (2015) 'U-Net: convolutional networks for biomedical image segmentation', in Navab, N. (ed.) MICCAI 2015. Munich: Springer, pp. 234–241.

SECTION 04

AMA & NLM Styles

American Medical Association | National Library of Medicine

For JAMA, AMA Journals & PubMed-Indexed Sources

AMA STYLE

- Superscript numbers in text (same as Vancouver)
- Reference list in order of citation
- All authors listed (no et al. limit in reference list)
- Abbreviated journal names (NLM format)
- Year;Volume(Issue):Pages — no spaces

Example:
Gulshan V, Peng L, Coram M, et al. Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. JAMA. 2016;316(22):2402-2410. doi:10.1001/jama.2016.17216

NLM / ICMJE STYLE

- Used by PubMed, Medline, ICMJE journals
- Virtually identical to Vancouver
- Adds PMID (PubMed ID) at end: PMID: 12345678
- PubMed Central ID if available: PMCID: PMC12345

Example:
Rajpurkar P, Irvin J, Zhu K, et al. CheXNet: radiologist-level pneumonia detection on chest X-rays with deep learning. arXiv. 2017;arXiv:1711.05225. PMID: 29182711

Side-by-Side Comparison of Major Styles

Feature	Vancouver	APA 7th	Harvard	AMA
In-text format	Superscript[1]	(Author, Year)	(Author, Year)	Superscript[1]
Citation order	Appearance	Alphabetical	Alphabetical	Appearance
Author limit	6, then et al.	2, then et al. from 3+	3, then et al.	All authors
Journal name	Abbreviated	Full, italic	Full or abbrev.	Abbreviated
Title case	Sentence	Sentence	Title Case	Title Case
Used in	BMJ, Lancet, IJMR	Psychiatry, MPH	Pub Health essays	JAMA, AMA
Software support	Zotero, Mendeley	Zotero, Mendeley	Zotero	Zotero, EndNote

SECTION 05

In-Text Citations

How to Cite Within Your Manuscript or Thesis

Placing Citations Correctly in Manuscript Text

Single Author — Vancouver

✗ Artificial intelligence "is transforming medicine" (Topol 2019)

✓ Artificial intelligence is transforming medicine.[1] OR: Topol[1] argued that artificial intelligence is transforming medicine.

Multiple Authors — APA

✗ LeCun, Bengio, and Hinton (2015) described deep learning in their study...

✓ LeCun et al. (2015) described deep learning... [3+ authors use et al. from first citation in APA 7th]

Direct Quote — Harvard

✗ Topol said 'AI will replace radiologists'

✓ "AI will replace radiologists" (Topol, 2019, p. 44) — must include page number for direct quotes in Harvard

Secondary Citation

✗ Hinton (cited in Goodfellow, 2016) stated...

✓ Use primary source whenever possible. If unavoidable: (Hinton, 1986, as cited in Goodfellow et al., 2016) — cite both

Digital Identifiers — The Modern Standard

DOI (Digital Object Identifier)

Unique persistent identifier assigned to articles.
Vancouver: doi:10.1001/jama.2016.17216
APA: https://doi.org/10.1001/jama.2016.17216
Always check CrossRef.org if DOI unknown

PMID (PubMed ID)

NLM/ICMJE format appends PMID at end.
Example: PMID: 27898976
Locate at: https://pubmed.ncbi.nlm.nih.gov/
Essential for biomedical references

PMCID (PubMed Central ID)

Identifies free full-text articles in PMC.
Example: PMCID: PMC5457749
Required by NIH grant-funded publications
Used alongside PMID in NLM style

URLs & Websites

Include access date for all web sources.
Vancouver: [cited 2024 Jan 10]
APA: (Retrieved January 10, 2024, from...)
Harvard: (Accessed: 10 January 2024)

arXiv / Preprints

Always label as [Preprint]
Format: arXiv:YYMM.NNNNN
Example: arXiv:2010.11929
Note limitations — not peer reviewed

ORCID

Unique author identifier (not in reference)
Helps disambiguation of author names
Add to manuscript metadata & cover letter
https://orcid.org/0000-0000-0000-0000

Tools Every PG Student Must Use

Zotero

FREE & OPEN SOURCE

Browser plugin, Word/LibreOffice integration, group libraries, PDF annotation, automatic DOI import

Limitation: Cloud storage limited (300MB free)

Best for: ALL PG students — FREE, powerful, actively developed

Mendeley

FREE (Elsevier)

14GB cloud storage, PDF manager, social network features, robust Android/iOS apps

Limitation: Elsevier integration may restrict data; some features require paid Elsevier access

Best for: Students wanting PDF cloud storage and mobile access

EndNote

PAID (Clarivate)

Industry gold standard, journal submission integration, 7000+ citation styles

Limitation: Expensive (~$300); steep learning curve; overkill for most PG students

Best for: Senior researchers, faculty, those submitting to high-impact journals

RefWorks

INSTITUTIONAL

Cloud-based, Google Docs integration, easy to learn

Limitation: Requires institutional subscription; less feature-rich than Zotero/Mendeley

Best for: Institutions with site license

How Deep Learning Transforms Literature Management

Semantic Scholar

AI Tool

Powered by AI2 & NLP Models

AI-generated TLDR summaries, citation impact scoring, auto-extracts methods/results sections from PDFs

Use: Finding key DL papers, tracking citation networks

Elicit.org

AI Tool

LLM-based Research Assistant

Answers research questions by scanning papers, extracts relevant columns (sample size, methods, outcomes, limitations)

Use: Systematic review screening, gap analysis

Scite.ai

AI Tool

Smart Citation Analysis

Deep learning model classifies each citation as 'supporting', 'contrasting' or 'mentioning' — shows evidence weight

Use: Identifying contested evidence, citation quality assessment

Connected Papers

AI Tool

Graph Neural Network Mapping

Builds visual similarity graph of papers using embedding-based NLP — not just citation links

Use: Discovering related DL papers outside your usual search terms

ResearchRabbit

AI Tool

AI Recommendation Engine

Continuously suggests new relevant papers based on your library; tracks new publications automatically

Use: Staying current with fast-moving DL/AI medical literature

ChatGPT / Claude

AI Tool

Large Language Models

Can draft reference lists, suggest citation styles, identify missing fields — ALWAYS verify output independently

Use: First-pass formatting; NOT reliable for exact bibliographic data

SECTION 07

Common Errors

Mistakes That Cost Marks & Credibility

Top 8 Mistakes Made by PG Students

✗ Wrong style for journal

HIGH

Fix: Always check the journal's 'Instructions to Authors' page before submission. Most Indian journals use Vancouver.

✗ et al. used incorrectly

HIGH

Fix: Vancouver: >6 authors; APA 7th: 3+ authors from first cite; AMA: do NOT use et al. in reference list.

✗ Journal not abbreviated

MED

Fix: Use NLM Catalogue at ncbi.nlm.nih.gov/nlmcatalog to find official abbreviations.

✗ DOI missing or wrong format

MED

Fix: Use CrossRef.org to verify DOIs. APA uses 'https://doi.org/'; Vancouver uses 'doi:'.

✗ Page numbers incomplete

MED

Fix: Use full pages (234-241) or acceptable abbreviation (234-41) per style guide. Never 234.

✗ Access date omitted for websites

MED

Fix: All web sources need accessed/cited date. Use: [cited 2024 Jan 10] (Vancouver) or (Retrieved Jan 10, 2024).

✗ Title capitalisation wrong

LOW

Fix: APA/NLM: sentence case. AMA/Vancouver: title case sometimes. Check style guide precisely.

✗ Auto-generated reference wrong

CRITICAL

Fix: ALWAYS manually verify Zotero/Mendeley/AI-generated references against original source.

Unique Challenges When Citing AI Research

Citing a GitHub Repository / Code	Vancouver: Chollet F. Keras [Source code]. GitHub; 2015 [cited 2024 Jan]. Available from: https://github.com/fchollet/keras APA: Chollet, F. (2015). Keras [Computer software]. https://github.com/fchollet/keras
Citing a Model / Dataset Version	Include version number and DOI if available. Example: OpenAI. (2023). GPT-4 technical report (Version 1.0) [Technical report]. https://arxiv.org/abs/2303.08774 Zenodo datasets have DOIs — always use them.
Citing a Retracted Paper	If you must cite a retracted paper (for historical context): Shame H, et al. [Retracted] Deep learning for diagnosis. J Med AI. 2020;5:12. Retraction in: J Med AI. 2021;6:1. Always check PubMed retraction status before citing.
Citing AI-Generated Content (LLMs)	Most journals do NOT allow citing ChatGPT/Claude as a reference. Use LLMs as a tool, not a citable source. If unavoidable — APA: OpenAI. (2024). ChatGPT (Mar 2024 version) [Large language model]. https://chat.openai.com
Pre-published / 'Ahead of Print'	Vancouver: Sharma R, et al. DL in cardiac imaging. Indian Heart J. Forthcoming 2024. doi:10.xxxx/ihj.2024.xxx APA: Sharma, R., et al. (in press). DL in cardiac imaging. Indian Heart Journal.

SECTION 08

Thesis & Dissertation

Referencing for MD / MS / DNB Theses

NMC / University Requirements for Thesis Referencing

THESIS REFERENCE CHECKLIST

- ☐ Style confirmed with guide / university
- ☐ All in-text citations have reference entry
- ☐ All reference entries cited in text
- ☐ Journal names correctly abbreviated
- ☐ Author names formatted consistently
- ☐ DOIs included for all journal articles
- ☐ Website access dates recorded
- ☐ No duplicate citation numbers (Vancouver)
- ☐ Reference management software used
- ☐ Supervisor has reviewed reference list
- ☐ Plagiarism check completed (Turnitin/iThenticate)
- ☐ Final proofread against style guide

TOP THESIS REFERENCING FAILURES

Using website-only sources for >30% of references

Thesis may be referred back

Citing textbooks for facts available in primary literature

Examiner flags as shallow review

Inconsistent style (mixing Vancouver & APA)

Automatic deduction of marks

Missing DOIs in journal references

Flagged by plagiarism software

Auto-import from Zotero without verification

Wrong years, authors or titles

Citing papers not actually read (honorary citations)

Academic misconduct risk

Reference list not matching bibliography

Common error in Indian MD theses

Key Takeaways

1. **Vancouver is king in biomedical journals**
   Master it first — it is used by Lancet, NEJM, BMJ, Indian J Med Res and most NMC thesis requirements.

2. **Always check the journal's style guide**
   No universal rule — check Instructions to Authors before every submission.

3. **Use Zotero — it's FREE and powerful**
   Import papers directly from PubMed and DOI. Export to any style in seconds. Eliminate manual errors.

4. **AI tools aid but never replace verification**
   ChatGPT/Zotero auto-import can hallucinate or mis-format. Always verify against the original source.

5. **DL/AI papers have special referencing needs**
   arXiv preprints, GitHub repos, model versions, and datasets each have specific citation formats.

6. **Correct referencing is ethical scholarship**
   Every incorrect citation is a disservice to the author — and risks plagiarism flags in your thesis.

Quick Reference Card

Vancouver	Superscript, numbered, abbrev journal
APA 7th	(Author, Year), alpha order, DOI link
Harvard	(Author, Year), title case, full journal
AMA	Superscript, all authors, JAMA format
NLM	Vancouver + PMID/PMCID
Zotero	Free, Word plugin, all styles
Elicit	AI gap & reference finder